

Zhuoqun (Carol) Wang

zhuoqun.wang0120@gmail.com | +1 410-350-9297 | WeChat: WangZhuoqun120 | San Francisco Bay Area

[LinkedIn](#) | [Google Scholar](#)

SUMMARY

Statistics PhD and Meta data scientist working at the intersection of tabular generative modeling, complex domain data, evaluation systems, and AI-assisted analytics workflows. PhD research developed tree-based generative models and contextualized generation methods for tabular data, with applications and evaluation on microbiome compositional / 16S rRNA sequencing data. Industry work supported identity-linkage/account-deduplication systems through noisy-label evaluation, people-level measurement, structured semantic metadata, monitoring, and analytics-agent evaluation.

FOCUS AREAS

Tabular generative modeling and contextualized generation for scientific data • Microbiome compositional / 16S rRNA sequencing data • Identity linkage and account deduplication • Noisy-label evaluation and label-quality monitoring • Structured semantic metadata • LLM-assisted analytics and agent evaluation

WORK EXPERIENCE

Meta Platforms, Inc.

Data Scientist

June 2024 – Present

Menlo Park, CA

- Implemented structured semantic metadata for identity-linkage analytics domains, encoding data assets, label semantics, cross-table logic, model/version caveats, and tacit labeling-process knowledge into agent-readable context; designed baseline analytics-agent evaluations showing over-80% pass rates across three tested domains.
- Supported evaluation of an existing identity-linkage/account-deduplication framework for externally reported people-level activity metrics, translating noisy multi-account behavior and metric anomalies into root-cause analysis, release evidence, and stakeholder communications.
- Investigated noisy-label and label-quality issues in identity-linkage training/evaluation data; designed monitoring metrics and built a continuous pipeline focused on label quality and model performance for a model supporting external people-level reporting.
- Drove transition from expert-dependent label evaluation to scalable vendor-labeling workflows, preserving measurement quality while delivering approximately \$350K OpEx savings per half.
- Used identity-linkage data to distinguish account-level artifacts from people-level behavior in product experiments, clarifying cases where people-level readouts changed or softened launch interpretations.
- Built AI-assisted reporting and RCA workflows, including reusable investigation playbooks for recurring model precision/recall/F1 regressions.

Meta Platforms, Inc.

Data Scientist Intern, Ads Infrastructure

May 2022 – Aug 2022

Menlo Park, CA

- Developed time-series forecasting and validation workflows for compute quota allocation in high-variance infrastructure systems, reducing under-utilization risk and informing allocation decisions.
- Built exploratory analyses and dashboards for infrastructure usage patterns, communicating findings to cross-functional partners and shaping follow-up modeling direction.

SELECTED RESEARCH

Tree-based generative models for tabular data with applications in microbiome compositional data

PhD dissertation, Duke University

- Developed tree-based generative/statistical models for sparse, high-dimensional tabular and compositional data, with applications to microbiome / 16S rRNA sequencing studies.
- Studied contextualized generation for scientific tabular data, comparing parametric Bayesian mixed-effects models, conditional tree flows, and marginal tree flows with Langevin posterior sampling on real microbiome data.
- Evaluated generation quality and tradeoffs across contextualized generation methods on real microbiome data, producing early empirical guidance on method behavior for controllable and conditional generation in structured tabular domains.

Generative modeling of conditional densities through additive tree flows

arXiv preprint

- Developed additive tree-flow methods for conditional density modeling and controllable generation in tabular data settings; released as [arXiv:2406.05260](#).

SELECTED PUBLICATIONS

- Wang Z**, Mao J, and Ma L. **A tree-based model for addressing sparsity and taxa covariance in microbiome compositional count data.** *Statistics in Medicine*. Accepted.
- Wang Z**, Awaya N, and Ma L. **Generative modeling of conditional densities through additive tree flows.** [arXiv:2406.05260](#).
- Siddiqui NY, Ma L, Brubaker L, Mao J, Hoffman C, Dahl EM, **Wang Z**, Karstens L. **Updating Urinary Microbiome Analyses to Enhance Biologic Interpretation.** *Frontiers in Cellular and Infection Microbiology*, 2022.
- Wang Z** and Durrett R. **Extrapolating weak selection in evolutionary games.** *Journal of Mathematical Biology*, 2019.

EDUCATION

Duke University – PhD, Department of Statistical Science; GPA: 3.86/4.00

Durham, NC | Aug 2019 – Dec 2024

Duke University – M.S., Department of Statistical Science

Durham, NC | Aug 2019 – Dec 2022

Nanjing University – B.S., Statistics, Kuang Yaming Honors School; GPA: 4.77/5.00; Graduated with Distinction

Nanjing, China | Sep 2015 – Jul 2019

SKILLS

Programming & Data: Python, SQL, R, PyTorch, scikit-learn, pandas, NumPy

Modeling & Evaluation: tabular generative modeling, contextualized generation, tree-based models, statistical modeling, noisy-label evaluation, identity linkage, account deduplication, people-level measurement

AI-Native Workflows: structured semantic metadata, LLM-assisted analytics, analytics-agent evaluation, monitoring pipelines, RCA workflow automation

Scientific Data: microbiome compositional data, 16S rRNA sequencing data, scientific tabular data

SELECTED HONORS

ISBA World Meeting Student/Postdoc Best Paper Award • Duke Fall Trading Competition 1st Place • Duke Summer Research Fellowship • COMAP ICM Meritorious Winner